

Strategies for Executing Analytics Projects: Toward a Unified Framework of Methodologies

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ABSTRACT

Analytics-project methodologies remain fragmented across technical, managerial, and governance perspectives. This study systematically identifies and analyzes 18 methodological documents representing analytics-project methodologies, extensions, and comparative syntheses to develop PRODIG8, a unified model for analytics-project execution. The model was derived through iterative structural reconciliation, sequentially comparing methodologies and modifying the evolving architecture by adding, renaming, merging, or splitting methodological functions. The 18 methodologies were then remapped to the resulting structure and their recommendations consolidated. PRODIG8 comprises eight methodological dimensions organized into three functional components: six core execution dimensions, Governance and Ethics as a transversal control dimension, and Continuous Improvement as an adaptive feedback dimension. This execution–control–adaptation architecture integrates technical execution, organizational governance, and continuous learning within a single model. PRODIG8 provides a systematically derived analytical structure for organizing, comparing, and refining analytics-project practices while preserving adaptability across organizational contexts.

Keywords

Business Analytics; Methodology; Governance; CRISP-DM; TDSP; Agile Analytics; Continuous Improvement.

1 Introduction

Over recent decades, the digitalization of processes and the expansion of analytical capabilities have profoundly reshaped how organizations operate, compete, and make decisions (Davenport & Harris, 2007; Provost & Fawcett, 2013). In this dynamic environment, the ability to collect, analyze, and interpret large volumes of information has become a strategic asset, enabling firms to optimize processes, anticipate market trends, and foster innovation (Davenport & Harris, 2007; Provost & Fawcett, 2013). Accordingly, analytics is no longer an auxiliary function but a central driver of value creation and long-term competitiveness (Provost & Fawcett, 2013). The increasing development of business analytics solutions within organizations reflects the exponential growth, diversity, and complexity of data generated by digital transformation, the Internet of Things (IoT), Cyber-Physical Systems (CPS), and advances in artificial intelligence (AI) and machine learning (ML) (Azeroual et al., 2025; Bemthuis et al., 2025; Fayyad et al., 2000; Huber et al., 2019; Ma et al., 2025).

Despite technological advances, the organizational landscape of analytics adoption remains complex. Many initiatives fail to deliver measurable value (Martinez-Plumed et al., 2021) due to persistent barriers, including heterogeneous data sources, data quality management, governance, privacy, and compliance with evolving legal standards (Azeroual et al., 2025; Ma et al., 2025; Montoya-Murillo et al., 2025; Wirth & Hipp, 2000). Recent studies report that approximately 87% of data science projects never reach production, and that over 80% of organizations lack a formal methodology for analytics execution (Saltz & Krasteva, 2022). Moreover, analytics projects are hindered by socio-technical and managerial obstacles, including siloed coordination among multidisciplinary teams, weak change-management practices, and limited stakeholder engagement (Bokrantz et al., 2024; Hamdani et al., 2024; Petrin et al., 2025; Zolbanin & Aubert, 2025). These issues are compounded by the misalignment between rapidly expanding data capabilities and organizations' analytical maturity, leading to underutilization of valuable data and fragmented project execution (Azeroual et al., 2025; Fayyad et al., 2000).

The diversity of project contexts and disciplinary origins has produced a wide range of methodological frameworks—such as KDD (Angée et al., 2018; Martinez et al., 2021; Saltz & Krasteva, 2022; Schröer et al., 2021), CRISP-DM (Fayyad et al., 1996a; Matignon, 2007; Wirth & Hipp, 2000, 2000), and SEMMA (*CRISP-DM 1.0*, 2000; Fayyad et al., 1996b; Schröer et al., 2021)—each valuable yet limited in adaptability, contextualization, and governance. In response, newer frameworks, including ASUM-DM (Agrawal & Shafer, 1996), TDSP (Microsoft Corporation, 2016), CASP-DM (Martinez-Plumed et al., 2021), and OSEMN (Bhardwaj, 2020), have emerged, incorporating agile and project-management principles. Complementing these, the INFORMS Analytics Process (INFORMS, 2014) provides a professional standard of practice widely adopted in industry, outlining seven stages—from business problem framing to lifecycle management—that link analytical rigor with decision-making and governance. The Data Mining Methodology for Engineering Applications (DMME) (Huber et al., 2019) further extends CRISP-DM by adding three technical phases—data acquisition, sensor integration, and engineering experimentation—that link measurement systems and analytical modeling in cyber-physical production environments. Hybrid implementations, such as those combining CRISP-DM with agile Scrum practices in supply-chain analytics (Hamdani et al., 2024), demonstrate how iterative sprints and user feedback loops improve responsiveness, team coordination, and socio-technical integration within complex analytics projects. In small and medium-sized enterprises, where analytical teams and technical resources are typically limited, agile approaches such as Scrum and XP offer lightweight, iterative alternatives that enhance adaptability, stakeholder communication, and incremental value delivery within analytics projects (Salazar-Salazar et al., 2024). Nevertheless, methodological fragmentation persists; studies often analyze these models in isolation, rarely comparing or integrating them (Fois et al., 2020; SAS Institute Inc., 2018). This lack of consolidation results in inefficiencies, inconsistent project execution, and reduced replicability of outcomes (Angée et al., 2018; Fois et al., 2020; Martinez-Plumed et al., 2021; Montoya-Murillo et al., 2025; Plotnikova et al., 2020; Salazar-Salazar et al., 2024). Therefore, an integrative synthesis is required to unify these dispersed approaches and provide a comprehensive methodological foundation for the execution of analytics projects.

This study addresses this fragmentation by systematically identifying and integrating methodologies for executing analytics projects and deriving, from their cumulative methodological contributions, a unified analytical model of analytics-project execution. Unlike prior reviews that primarily compare or classify existing approaches (Martinez et al., 2021; Saltz & Krasteva, 2022), the study reconstructs their methodological lineage and examines how successive frameworks progressively differentiate and extend the functions required for project execution. This evolution encompasses advances in project management and agility (Hamdani et al., 2024; Salazar-Salazar et al., 2024), domain-specific technical requirements (Huber et al., 2019), governance and organizational integration (Montoya-Murillo et al., 2025), and iterative learning and operational adaptation (Bokrantz et al., 2024). Recent adaptations of

CRISP-DM also demonstrate how analytics methodologies can extend beyond conventional business applications to support institutional decision-making and governance (Herrera Franco et al., 2025). By sequentially reconciling the identified methodologies and consolidating their recommendations, this study develops PRODIG8. The resulting model comprises eight methodological dimensions organized into three distinct functional roles: six dimensions form the core execution lifecycle, Governance and Ethics provide transversal control, and Continuous Improvement provides adaptive feedback across the lifecycle. PRODIG8 therefore integrates technical, managerial, governance, and learning requirements within a common execution–control–adaptation architecture for analytics-project management.

By consolidating methodological advances and aligning them with recognized standards such as ISO/IEC 29110 (Montoya-Murillo et al., 2025), this study contributes to the methodological maturity and standardization of business analytics. More specifically, PRODIG8 provides a systematically derived analytical architecture for organizing, comparing, and refining analytics-project practices while preserving the adaptability required across organizational contexts. Its distinction among execution, transversal control, and adaptive feedback provides a common structure for examining both the completeness of project methodologies and the relationships among their technical, managerial, governance, and learning functions. The remainder of this paper is organized as follows: Section 2 presents the theoretical framework; Section 3 describes the methodological design and analytical construction procedure; Section 4 presents the resulting PRODIG8 architecture and the consolidated methodological recommendations associated with its dimensions; Section 5 discusses its theoretical and practical implications; Section 6 outlines the study’s limitations; and Section 7 concludes.

Before presenting the methodological workflow, it is essential to situate the unified framework within the broader theoretical lineage of analytics methodologies. The following section, therefore, reviews the main paradigms that have shaped the execution of analytics projects, tracing how successive generations of process models have progressively integrated governance, ethics, and iterative learning.

2 Theoretical Framework

The execution of analytics projects has evolved through successive methodological paradigms, closely mirroring technological progress and organizational transformation. Each paradigm reflects the technological and organizational conditions of its time. Foundational models such as Knowledge Discovery in Databases (KDD) (Fayyad et al., 1996b), CRISP-DM (Chapman et al., 2000), and SEMMA (SAS Institute Inc., 1998) established the procedural basis for structured data mining and analytics practice. These frameworks share a sequential logic—progressing from problem definition and data preparation to modeling, evaluation, and deployment—that provided early standardization for analytical workflows. Their principal contribution lies in defining reproducible stages and deliverables, yet their tool-oriented design and limited attention to governance and adaptability restrict their suitability for contemporary, dynamic environments. Despite these limitations, they remain the conceptual backbone of modern analytics methodologies, establishing the sequential logic and standardized deliverables that later frameworks—such as ASUM-DM, TDSP, DMME, and DataPro—would refine through iterative, agile, and governance-oriented extensions. Beyond its structural clarity, CRISP-DM remains historically significant as the first methodology to formalize a standard, repeatable, and technology-agnostic process model for data mining, introducing iterative feedback loops and documentation practices that continue to underpin subsequent frameworks, including those integrated in this study’s unified model. In addition, domain-specific adaptations, particularly in education and academic management, have expanded CRISP-DM through process mining integrations that enable curriculum evaluation and predictive student performance analytics, demonstrating its cross-domain versatility (Herrera Franco et al., 2025).

Recent adaptations have further evolved CRISP-DM toward quality assurance and lifecycle governance. Notably, CRISP-ML(Q) (Studer et al., 2021) extends CRISP-DM by incorporating explicit quality management, reproducibility, and long-term model maintenance steps, reflecting an increasing convergence between methodological rigor and operational reliability. These extensions echo the same need for continuous monitoring and improvement that underpins the eighth phase of the unified framework proposed in this study.

Subsequent frameworks, such as ASUM-DM (IBM, 2015) and TDSP (Microsoft Corporation, 2016), expanded on these classical models by integrating principles from software engineering and agile project management. ASUM-DM introduced lifecycle documentation, stakeholder engagement mechanisms, and process iteration, while TDSP emphasized automation, version control, and continuous integration. Developed by IBM as an evolution of CRISP-DM, ASUM-DM explicitly integrates project and communication management activities to coordinate multi-organizational analytics initiatives and to align technical execution with governance and stakeholder requirements.

Similarly, TDSP aligns analytical development with DevOps principles, emphasizing automation and reproducibility across collaborative teams. Recent hybrid methodologies reinforce that combining CRISP-DM's structured phases with agile's iterative and human-centered practices enhances collaboration, responsiveness, and alignment between technical execution and organizational learning (Hamdani et al., 2024). Agile-oriented methodologies typically emphasize seven recurring attributes—flexibility, responsiveness, speed, lean development, simplicity, lightweight documentation, and limited formal deliverables—which underpin the adaptive behavior now integrated into modern analytics frameworks.

Complementing these research- and vendor-oriented methodologies, the INFORMS Analytics Process (INFORMS, 2014) represents a practitioner-standard framework endorsed by the Institute for Operations Research and the Management Sciences. It structures analytics projects into seven sequential yet iterative stages—business problem framing, analytics problem framing, data, methodology, model building, deployment, and lifecycle management—providing a governance-oriented bridge between business objectives and technical execution. Designed originally for professional certification under the Certified Analytics Professional (CAP) program, the INFORMS framework operationalizes communication, accountability, and value realization across project phases. Its emphasis on traceability and stakeholder alignment directly supports the governance-aware orientation of the unified framework proposed in this study.

Similarly, methodologies such as DMME, MAISTRO, and DataPro advanced socio-technical considerations, embedding elements of collaboration, ethics, and organizational governance into the analytics lifecycle. DMME (Huber et al., 2019), in particular, extends CRISP-DM by introducing three additional phases—Technical Understanding, Technical Realization, and Technical Implementation—that explicitly address data acquisition, sensor integration, and process alignment in engineering environments. These three phases bridge analytical modeling with real-time data acquisition, connecting business objectives to operational feasibility in industrial and cyber-physical contexts. Each phase corresponds to a specific gap in CRISP-DM: Technical Understanding adds requirements for parameter definition and experimental planning; Technical Realization formalizes sensor installation and data collection procedures; and Technical Implementation ensures that acquired data flows into operational analytics systems through runtime integration and validation. This layered extension was empirically validated in industrial testbeds and pilot projects, confirming its applicability for integrating analytical models within engineering and production systems (Huber et al., 2019). Collectively, these frameworks illustrate a methodological shift from static, sequential pipelines toward dynamic, iterative, and governance-aware lifecycles that reflect the growing complexity and interdisciplinarity of analytics projects. Despite these advances, the literature continues to report persistent challenges, including methodological fragmentation, weak interoperability among process models, and insufficient standardization of roles, responsibilities, and quality metrics (Martinez et al., 2021; Saltz & Krasteva, 2022).

Within this evolving landscape, analyses such as Shimaoka et al. (2024) highlight the proliferation of domain-specific extensions that lack a unifying methodological standard, reinforcing the need for consolidation and cross-domain integration. Their findings reveal that while specialized frameworks have advanced practice within individual sectors, the absence of a holistic structure limits interoperability and knowledge transfer across domains—precisely the gap this study seeks to address.

Comparative reviews show that these frameworks converge conceptually around five foundational principles: (1) structured phases that translate business problems into analytical tasks; (2) iteration and feedback as mechanisms for learning; (3) cross-functional collaboration; (4) emphasis on data quality and reproducibility; and (5) increasing concern for ethical and governance dimensions. Nonetheless, divergences persist in phase granularity, governance integration, and treatment of post-deployment activities. Some models (e.g., CRISP-DM, SEMMA) end at deployment, whereas newer approaches (e.g., TDSP, MAISTRO) extend to continuous operation and maintenance. Others differ in their approach to ethics, transparency, and accountability—dimensions that are often external to early process models but are now central to responsible analytics practice. OSEM, widely adopted in real-time and applied data science contexts, exemplifies the operational simplicity and adaptability that many modern methodologies, including the unified framework proposed here, seek to institutionalize.

Across these successive paradigms, the methodological landscape reveals both conceptual continuity and fragmentation. Existing reviews (Martinez et al., 2021; Saltz & Krasteva, 2022) underscore the absence of a comprehensive framework that consolidates methodological diversity while embedding governance, adaptability, and standardization. Theoretical gaps remain concerning (a) the integration of governance and ethics across all phases; (b) the formalization of iterative learning cycles that connect technical and managerial processes; and (c) the definition of cross-phase feedback mechanisms ensuring methodological coherence. Addressing these gaps requires reconceptualizing analytics methodologies not as fixed pipelines but as adaptive systems that unify technical, managerial, and ethical layers within a single structure.

The model proposed in this study responds directly to these theoretical gaps by integrating agile practices with structured lifecycle approaches, consistent with recent calls for hybrid methodologies in analytics execution (Saltz & Krasteva, 2022). It consolidates the principal methodological advances identified across successive frameworks—standardization, iterative development, operationalization, governance, and continuous learning—within a differentiated architecture. In this architecture, the core execution dimensions provide the structured lifecycle required for systematic project development; Governance and Ethics operates as a transversal control mechanism; and Continuous Improvement provides the adaptive feedback mechanism through which results, operational experience, and changing conditions can reshape execution activities. This execution–control–adaptation structure aligns with the growing emphasis on methodological maturity, accountability, and ethical responsibility reflected in standards such as ISO/IEC 29110 and IEEE 7000. PRODIG8 therefore conceptualizes analytics-project management as an integrated socio-technical system in which structure, control, and adaptation perform distinct but interdependent functions.

3 Materials and methods

This section outlines the workflow adopted for the systematic literature analysis, following established methodological frameworks (Aria & Cuccurullo, 2017; Donthu et al., 2021; Page et al., 2021). The procedure is structured into three stages: Section 2.1 outlines the study design, Section 2.2 describes data collection and preparation, and Section 2.3 details the analytical construction of PRODIG8. All procedures, inclusion criteria, and coding schemes were documented in structured spreadsheets to ensure transparency and replicability. The citation-expansion stage identified the methodological corpus and traced its lineage, while the subsequent sequential structural-reconciliation procedure derived the unified architecture. Once the final structure was established, the recommendations of all 18 methodologies were remapped and consolidated within its dimensions. This separation between corpus identification, structural derivation, and recommendation consolidation provides the methodological basis for the PRODIG8 model.

3.1 Study design

The methodological design of this study followed the principles for systematic reviews proposed by Kitchenham (2004) and Page et al. (2021) and was grounded in an iterative, citation-based discovery strategy developed to trace the evolution of methodologies for executing analytics projects. Rather than relying on topic-based keyword searches, we employed a forward-citation expansion approach to identify methodological innovations by tracing their reference relationships to established frameworks. This approach is efficient for capturing new or derivative methodologies because it follows the field’s intellectual lineage, revealing conceptual extensions and refinements that may not share common terminology. Although this method may not capture independently developed methodologies lacking citation ties to foundational models, Saltz and Krasteva (2022) indicate that most contemporary frameworks are, in fact, derivations of classical models, suggesting broad coverage. By recursively tracing citation linkages, this design ensures completeness through saturation of methodological descendants. At the same time, reliability is strengthened by limiting inclusion to peer-reviewed frameworks that formalize process models rather than isolated applications.

The process began with four seminal seed methodologies representing the foundations of analytics project development: Knowledge Discovery in Databases (Fayyad et al., 1996a), Sample–Explore–Modify–Model–Assess (SEMMA; SAS Institute, 1998), Cross-Industry Standard Process for Data Mining (CRISP-DM; Chapman et al., 2000), and Analytics Solutions Unified Method for Data Mining (ASUM-DM; IBM, 2015). A forward-citation search was conducted in the Scopus database to retrieve all publications that cited any of these seed papers between 2010 and 2025.

Each retrieved document was screened to determine whether it proposed, formalized, or extended a methodological framework for analytics or data-science projects. Only those documents that introduced a genuinely new methodology or a substantive modification to an existing one were incorporated as additional seeds for the next iteration. This selective, recursive expansion ensured controlled growth of the seed set and avoided indiscriminate inclusion of citing documents. The process was repeated until saturation was reached—that is, when no additional unique methodologies appeared among the newly retrieved citing records. The parameters applied in the search process are summarized in **Table 1**, and the iterative expansion workflow is illustrated in **Fig. 1**.

It should be noted that the citation-expansion search might omit frameworks developed entirely independently of the classical methodological lineage—that is, those not citing any of the seed papers (KDD, SEMMA, CRISP-DM, or ASUM-DM). However, such cases appear rare. Saltz and Krasteva (2022) observed that the vast majority of recent analytics methodologies are direct extensions or modifications of these foundational models rather than entirely novel

approaches. Accordingly, the adopted strategy likely captured the principal frameworks proposed between 2010 and 2025. Future research could complement this approach by using targeted keyword searches to identify potential outliers and improve completeness.

Fig. 1 illustrates this selective citation-expansion process. The diagram begins with the four classical seed methodologies and shows successive forward-citation searches performed in Scopus. After each iteration, only validated methodological contributions (e.g., DMME, TDSP, MAISTRO) are added to the seed group, while non-methodological citing papers are excluded. The iterative loop continues until the final cycle yields no new methodological proposals, indicating conceptual saturation. The lower section of the figure shows the progressive refinement from the total number of retrieved records to the final corpus of 18 methodological documents retained for synthesis.

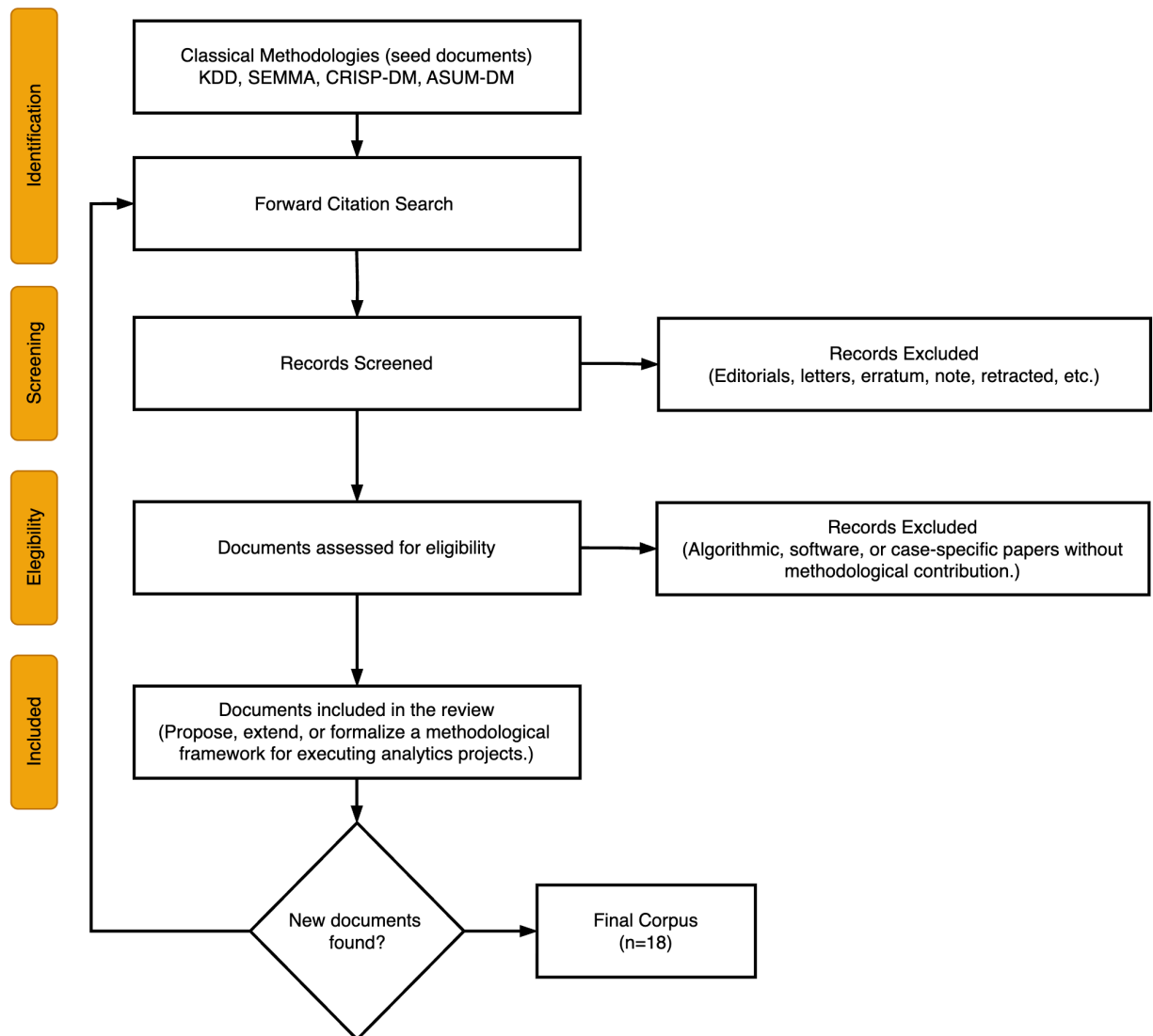


Fig. 1. Iterative citation-expansion process for identifying methodological frameworks. Only papers proposing new or modified frameworks were added as new seeds in subsequent iterations until conceptual saturation was reached.

Table 1.

Parameters of the Study.

Parameter	Value
Database	Scopus
Search type	Forward-citation search (iterative expansion)
Initial seed set	KDD (1996); SEMMA (1998); CRISP-DM (2000); ASUM-DM (2015)
Years of analysis	2010-2025
Language	English
Data retrieval	August 15, 2025
Document types	Peer-reviewed journal articles and conference papers
Inclusion criteria	Cites at least one seed or descendant methodology and proposes, extends, or formalizes a methodological framework for executing analytics projects
Exclusion criteria	Algorithmic, software, or case-specific papers without methodological contribution
Screening procedure	Manual screening of titles, abstracts, and full texts by two reviewers; discrepancies resolved by consensus
Quality assessment	Not applied

3.2 Data collection and preparation

After the final iteration, all records were consolidated and filtered according to the following criteria:

- (a) explicit citation of at least one seed or descendant methodology.
- (b) proposal or modification of a structured process for executing analytics projects; and
- (c) publication as a peer-reviewed journal article or conference paper.

Studies focusing solely on algorithmic innovation, software implementation, or isolated case applications without making methodological contributions were excluded.

The iterative process initially retrieved 699 records, which yielded a final corpus of 18 methodological documents after successive screening and consolidation. Two reviewers conducted Screening and classification independently, and discrepancies were resolved by consensus to ensure methodological reliability. Table 1 summarizes the parameters applied in the citation-based search, and Figure 1 visualizes the selective, iterative workflow described above.

All records retrieved through the iterative citation-based search were exported from Scopus in CSV format and consolidated into a single dataset. Duplicate entries were identified and removed, and the remaining documents were manually screened to ensure consistency with the inclusion criteria described in Section 3.1.

The final dataset comprised 18 methodological documents obtained through the selective citation-expansion process illustrated in Figure 1. For each methodology, a structured extraction sheet was used to record publication year, methodology name, domain of origin, the phases or stages defined by the source, the methodological recommendations associated with them, distinctive features, and references to methodological or governance standards (e.g., ISO/IEC 29110, IEEE 7000). At this stage, the terminology and organizational structure reported by each source were preserved rather than immediately standardized into a common phase structure.

Two reviewers extracted the data independently and then reconciled them through consensus to ensure consistency of the source-level information. The resulting records provided input for the sequential structural-reconciliation procedure described in Section 3.3 and, after the final PRODIG8 architecture was established, for the subsequent mapping and consolidation of methodological recommendations.

3.3 Data analysis

The 18 methodological documents were analyzed sequentially through an iterative structural-reconciliation procedure. The phases of the first document were used to establish an initial provisional structure for executing analytics projects. The methodological functions identified in each subsequent document were then compared with the current provisional structure. The phases proposed by each subsequent methodology were then compared with the current structure. At each iteration, the structure was retained when the incoming phases were already represented or modified when substantive differences were identified. Modifications involved adding phases that captured previously unrepresented functions, renaming existing phases to accommodate broader or more precise meanings, merging conceptually equivalent phases, or splitting phases when a methodology distinguished functions previously represented together. This procedure was repeated sequentially across the 18 methodologies, with each resulting

structure becoming the basis for comparison with the next methodology, until the final set of eight phases constituting PRODIG8 was obtained.

Once the final eight-phase structure was established, the 18 methodologies were reviewed again, and their methodological recommendations were mapped to the corresponding PRODIG8 phases. This second analytical stage used the final phase structure as a common reference framework, allowing recommendations expressed through different terminology, levels of granularity, or methodological organization to be systematically aligned. The recommendations assigned to each phase were subsequently compared and consolidated by integrating complementary contributions and eliminating conceptual redundancies while preserving distinctive methodological guidance. This process produced the consolidated content of each PRODIG8 phase and maintained traceability between the unified framework and the methodologies from which its recommendations were derived.

Two reviewers performed both analytical stages—the iterative construction of the phase structure and the subsequent mapping and consolidation of methodological recommendations—independently. The reviewers resolved differences in phase reconciliation, recommendation mapping, or interpretation through discussion and consensus to ensure consistency in constructing PRODIG8.

Four transformation operations governed structural reconciliation. Addition was applied when a methodology introduced a function not represented in the provisional structure. Renaming was applied when an existing phase represented the same underlying function but required a broader or more precise designation to accommodate the incoming methodological contribution. Merging was applied when phases represented separately in the provisional and incoming structures were determined to perform conceptually equivalent functions. Splitting was applied when an incoming methodology differentiated substantively distinct functions that were represented jointly in an existing phase. When none of these conditions applied, the existing phase structure was retained. These operations provided consistent decision rules for progressively reconciling heterogeneous methodological structures while preserving substantive distinctions among their project-execution functions.

3.4 Scope of Quality Appraisal

Because this study aimed to map and conceptualize methodological frameworks rather than evaluate empirical evidence, no formal quality appraisal was conducted. This decision aligns with established guidance on scoping and conceptual reviews, which aim to identify, synthesize, and conceptualize existing approaches rather than assess methodological rigor (Munn et al., 2018; Peters et al., 2020). The adopted approach ensures transparency and consistency with best practices for evidence synthesis in exploratory and theory-building research.

4 Results

This section presents the results of constructing PRODIG8 from the 18 selected methodologies. The sequential structural-reconciliation procedure progressively transformed the provisional phase structure as successive methodologies introduced equivalent, broader, narrower, or previously unrepresented functions. After reconciling all 18 methodologies, the procedure produced a stable architecture comprising eight interdependent phases. The complete corpus was then analyzed a second time, with each methodology's recommendations mapped to this final architecture and consolidated within the corresponding phases. The following subsections therefore present both the resulting phase structure and the methodological content assigned to each phase.

The resulting model, hereafter referred to as PRODIG8, comprises eight dimensions for structuring business analytics project execution. The term “PRODIG8” summarizes Project, Data, and Governance, reflecting its integration of project execution, data-centric rigor, and governance-oriented accountability. Rather than directly adopting any single methodology, the eight-phase architecture reflects the cumulative result of structural transformations identified across the 18 methodologies and the subsequent consolidation of their methodological recommendations.

Each phase is distinctly defined, with clear conceptual and procedural boundaries that ensure consistency and avoid overlap. The framework integrates DMME's logic by embedding its technical phases into data preparation and deployment activities, ensuring that sensor-based data collection and engineering context are explicitly addressed alongside governance and ethical oversight. In doing so, the unified model preserves the lineage of DMME's three technical phases—Technical Understanding, Technical Realization, and Technical Implementation—embedding them within the Data Preparation and Operation & Maintenance phases to maintain traceability between industrial engineering and business analytics contexts. Empirical demonstrations of DMME in vibration analysis and predictive maintenance testbeds confirm the feasibility and reproducibility of its phases for sensor-based data integration and

runtime deployment in manufacturing environments. For example, Data Preparation encompasses data transformation, integration, and quality enhancement, whereas Project Design focuses on selecting, training, and prototyping analytical models. Also, curriculum analytics studies based on CRISP-DM have successfully implemented the unified framework's iterative logic—combining data understanding, preparation, and process analysis—to identify dropout risks and optimize academic trajectories (Herrera Franco et al., 2025).

The sequential reconciliation also showed how the architecture of analytics-project methodologies expanded as new methodological concerns emerged. Earlier frameworks primarily established the core analytical workflow, whereas subsequent methodologies progressively differentiated project definition, technical and organizational activities, evaluation, operationalization, governance, and iterative learning. A phase was retained in the evolving structure when subsequent methodologies represented the same underlying function, while structural changes were introduced when a methodology distinguished a function that the existing architecture could not adequately represent.

As shown in Table 2, the sequential reconciliation progressively transformed the provisional structure through addition, renaming, merging, splitting, and retention operations. Once the eight-dimensional architecture emerged, subsequent methodologies were mapped against the evolving structure to test whether additional structural changes were required. This sequential reconciliation produced a stable architecture of eight distinct but interdependent methodological dimensions. Once this structure emerged, the remaining documents did not require additional dimensions; instead, they reinforced, refined, or validated its functional boundaries. These eight dimensions subsequently provided the reference structure for mapping and consolidating methodological recommendations.

Table 2.

Sequential structural reconciliation of the 18 methodologies and derivation of the PRODIG8 architecture.

Iteration	Methodology (year)	Methodological contribution to the evolving structure	Structural operation	Effect on provisional PRODIG8 structure
01	DMME (2019)	Integrates the conventional analytics workflow with explicit technical understanding, data-acquisition realization, and production-oriented technical implementation	Initialization	Establishes a nine-phase provisional structure spanning business understanding, technical planning and realization, data analysis, evaluation, technical implementation, and deployment
02	Improved ASUM-DM (2018)	Extends the technical analytics workflow toward application-level design and evaluation while introducing project management as a cross-cutting coordination function	Addition + merging	Adds Project Management as a transversal function; consolidates technical realization, modeling, infrastructure design, visualization, and prototyping into a broader Project Design function; retains business/data understanding, data preparation, evaluation, and deployment-related functions
03	MAISTRO (2025)	Extends the lifecycle beyond deployment to sustained operation and continuous adaptation, while embedding ethical governance, transparency, risk management, and accountability throughout AI-system development	Addition + renaming + merging	Introduces Governance and Ethics and Continuous Improvement as explicit transversal/adaptive functions; broadens Business Understanding to Project Scope Definition; consolidates technical implementation and deployment into Operation and Maintenance
04	OSEMN (2020)	Provides a compact data-science workflow centered on obtaining, scrubbing, exploring, modelling, and interpreting data; reinforces the technical analytical core while explicitly linking interpretation to the original business question and stakeholder-useful results	Mapping / reinforcement	Obtain → Data Understanding; Scrub → Data Preparation; Explore → Data Understanding; Model → Project Design; Interpret → Evaluation. No new dimension required
05	CRISP-DM (2000)	Reinforces the core analytics lifecycle from business definition through data work, modeling, evaluation, deployment, monitoring, and iterative learning	Retention	All CRISP-DM functions are representable within the existing eight dimensions; no structural modification is required

06	CRISP-DM-based ABS/process-mining methodology (2024)	Contextualizes CRISP-DM for specialized analytical projects by broadening business understanding to stakeholder/context analysis and data understanding to include analytical-tool capabilities; reinforces iterative modeling, evaluation, deployment, monitoring, and maintenance	Retention + refinement	Confirms the eight-dimensional structure; strengthens Project Scope Definition, Data Understanding, Project Design, Model Evaluation, Operation and Maintenance, and Continuous Improvement without requiring a new dimension
07	DataPro	Extends CRISP-DM with Technical Understanding and Implementation; explicitly separates business-to-technical translation from data understanding and modeling from real-world implementation; formalizes stakeholder/data-scientist responsibilities and feedback loops	Retention + refinement	Confirms the eight-dimensional structure while strengthening Project Scope Definition, Project Design, Operation and Maintenance, and Continuous Improvement; no new dimension required
08	Plotnikova et al. systematic review of data-mining methodologies (2020)	Synthesizes KDD, SEMMA, CRISP-DM and their adaptations; identifies recurrent technological and organizational extensions, particularly deployment/integration, project-management, organizational, quality-assurance, collaboration, feedback, and ongoing-support requirements	Triangulation + refinement	Provides cross-methodological validation of the eight-dimensional structure; particularly strengthens Project Scope Definition, Governance and Ethics, Operation and Maintenance, and Continuous Improvement; no new dimension required
09	Salazar-Salazar et al. — Agile SDLC review for BDAS (2024)	Compares CRISP-DM, TDSP, and DDSL with Scrum-XP using roles, phases/activities, and work products; incorporates agile project management, iterative development, product release, deployment, monitoring, maintenance, value evaluation, and improvement	Triangulation + refinement	Strongly validates Project Scope Definition, Project Design, Model Evaluation, Operation and Maintenance, and Continuous Improvement; strengthens the governance/project-management interpretation; no new dimension required
10	Agile SCRUM/CRISP-DM supply-chain analytics framework (2022)	Integrates CRISP-DM's analytical lifecycle with Scrum-based roles, sprints, backlog management, stakeholder participation, iterative replanning, feedback, risk handling, and learning	Integration + refinement	Confirms the eight-dimensional structure; substantially strengthens Governance and Ethics and Continuous Improvement as transversal/adaptive functions while refining Project Scope Definition and Project Design
11	Curriculum-analysis model integrating CRISP-DM, process mining and data analytics (2025)	Extends CRISP-DM with explicit project definition, data-management infrastructure, process analytics, KPI-based monitoring, optimization, and continuous feedback	Integration + refinement	Confirms the eight dimensions; strongly reinforces Project Scope Definition, Governance and Ethics, Operation and Maintenance, and Continuous Improvement; broadens interpretation of Project Design
12	KDD for Science Data Analysis (1996)	Shows that successful KDD requires domain-problem formulation, understanding of heterogeneous data, feature engineering/reduction, analytical-task and algorithm selection, validation, interpretability, scalability, implementation, and human-machine interaction	Validation + refinement	Confirms the eight dimensions; especially strengthens Data Understanding, Data Preparation, Project Design, Model Evaluation, and Operation and Maintenance; provides additional evidence for infrastructure as a cross-cutting concern rather than a separate dimension
13	Comparative heavyweight/lightweight BDAS SDLC review (2025)	Compares KDD, SEMMA, CRISP-DM, BDPL, ASUM, TDSP and DDSL against a software-engineering lifecycle covering project management and implementation; exposes systematic deficiencies in roles, activities, products and lifecycle completeness	Triangulation + boundary refinement	Strongly validates PSD, DU/DP, PD, ME and OM; provides major support for treating project management as embedded across PRODIG8; reinforces monitoring/maintenance and iteration; no evidence

				requiring a ninth execution dimension
14	CRISP-DM and predictive analytics framework for CRIS (2025)	Extends CRISP-DM toward predictive decision support by explicitly incorporating project scope, heterogeneous-data management, model optimization, operational integration, data governance, privacy, legal compliance, ethics, fairness auditing, and iterative model improvement	Retention + strong validation	Confirms all eight PRODIG8 dimensions; provides particularly strong validation for Governance and Ethics, Operation and Maintenance, and Continuous Improvement; no structural modification required
15	Enhanced CRISP-DM for AI in manufacturing (2024)	Extends CRISP-DM into a continuous AI-solution lifecycle by adding explicit Operation and Maintenance, incorporating early conceptual solution design, continuous monitoring of model/data drift, feedback-driven redesign, and cross-functional domain/data-science/data-engineering competence	Retention + strong refinement	Strongly validates Project Design, Operation and Maintenance, and Continuous Improvement; reinforces Project Scope Definition and Governance and Ethics as broader lifecycle concerns; no new dimension required
16	CRISP-DM + Scrum ML development guideline (2023)	Integrates Scrum ceremonies, artifacts, roles, iteration planning, stakeholder review, progress tracking, retrospectives, and deployment architecture into CRISP-DM	Integration + refinement	Confirms the eight-dimensional architecture; particularly strengthens Project Scope Definition, Governance and Ethics, Project Design, Model Evaluation, and Continuous Improvement; no new dimension required
17	MLPM for design-oriented ML research (2025)	Integrates CRISP-DM with DSR and ADR, explicitly separating problem identification from solution-objective formulation and adding design/development/refinement, evaluation, reflection/learning, and communication	Splitting challenge + retention/refinement	Confirms the eight-dimensional architecture overall; challenges Project Scope Definition by separating problem identification and solution objectives, strongly validates Project Design, Model Evaluation, and Continuous Improvement; Communication is treated as an output/activity rather than a new execution dimension
18	Comparative analysis of TDSP and ASUM-DM (2020)	Cross-method comparison identifies common and complementary lifecycle functions spanning business understanding, data acquisition/understanding, design, preparation, modeling, evaluation, deployment, operation, monitoring, optimization, governance, and iterative development	Triangulation + validation	Strongly validates the eight-dimensional structure; particularly reinforces the separation of Project Design, Operation and Maintenance, and Continuous Improvement; no new dimension required

The final iteration produced eight distinct but interdependent methodological dimensions, which constituted the stable structure used in the subsequent mapping and consolidation of methodological recommendations. Six dimensions represent the core execution lifecycle—Project Scope Definition, Data Understanding, Data Preparation, Project Design, Model Evaluation, and Operation and Maintenance—whereas Governance and Ethics operate as a transversal control dimension and Continuous Improvement functions as an adaptive feedback dimension connecting and renewing the lifecycle:

1. Project Scope Definition
2. Data Understanding
3. Data Preparation
4. Project Design
5. Model Evaluation
6. Governance and Ethics
7. Operation and Maintenance

8. Continuous Improvement

This iterative structure reinforces agile feedback loops observed in hybrid CRISP-DM applications, where regular evaluation and stakeholder interaction continuously refine objectives and deliverables (Hamdani et al., 2024). The resulting structure also aligns with agile lifecycle models such as Scrum-XP, reflecting iterative delivery, stakeholder feedback, and incremental value generation within the unified, governance-aware framework. The following subsections detail each phase, highlighting shared practices, distinctive contributions, and integrative guidelines.

Figure 2 presents the structural architecture of PRODIG8 as an iterative, governed, and adaptive model for executing business analytics projects. The architecture distinguishes six core execution dimensions—Project Scope Definition, Data Understanding, Data Preparation, Project Design, Model Evaluation, and Operation and Maintenance—connected through the primary execution flow. Governance and Ethics operates as a transversal control dimension, providing oversight, accountability, risk management, and ethical guidance across the execution lifecycle. Continuous Improvement operates as an adaptive feedback dimension, receiving information from the core execution dimensions and feeding learning and corrective actions back into them as project conditions, results, and operational experience evolve. Accordingly, PRODIG8 does not represent analytics-project execution as a strictly linear sequence, but as a controlled lifecycle in which execution, governance, and adaptive learning operate as interconnected mechanisms.

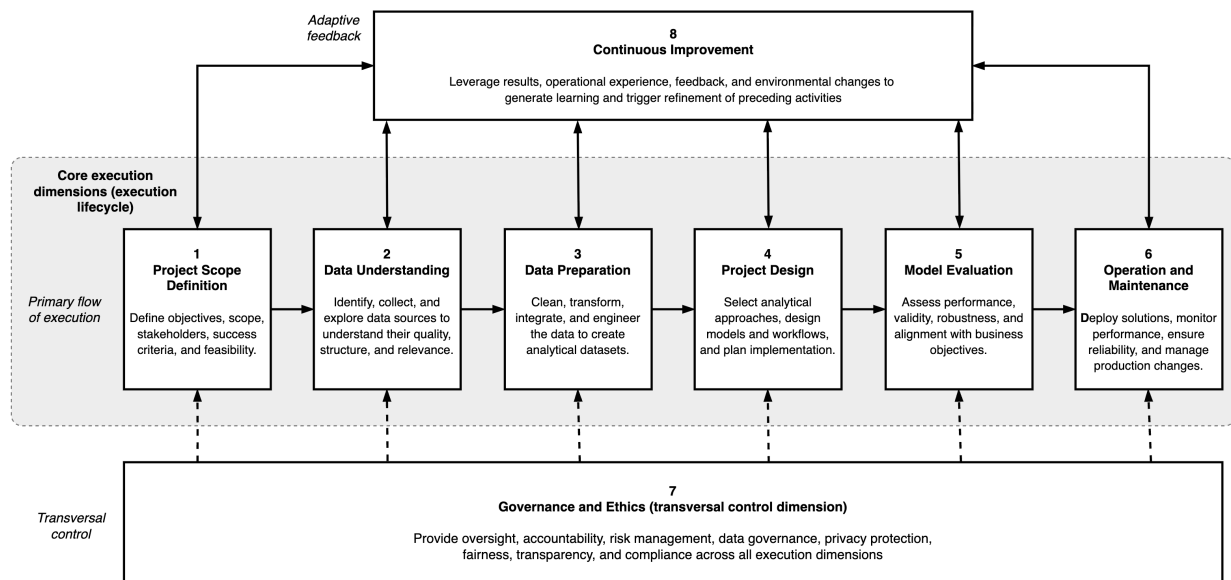


Figure 2. Structural architecture of the PRODIG8 model. The model integrates six core execution dimensions with a transversal Governance and Ethics dimension and an adaptive Continuous Improvement dimension.

4.1 Project Scope Definition

In this phase, methodologies converge on the principle that a business analytics project must always start with an explicit and multidimensional scope definition that integrates business needs, technical feasibility, and stakeholder alignment. The DMME framework insists that business objectives must be translated into measurable technical goals, directly linking them to system structures, processes, and operational constraints. This ensures that ambitions such as predictive maintenance or quality improvement are grounded in realistic parameters and technological limitations, creating a bridge between managerial priorities and engineering contexts (Huber et al., 2019). Similarly, other approaches emphasize the importance of beginning with a precise formulation of the research or analytical problem, including the selection of relevant variables, dimensions, and expected contributions. By framing the analysis in theoretical and empirical foundations, the scope gains both academic legitimacy and practical applicability (Angée et al., 2018). MAISTRO reinforces this vision by urging organizations to align artificial intelligence initiatives with strategic imperatives, focusing on genuine demands rather than speculative uses of technology. In this regard, the methodology prioritizes understanding user needs, evaluating feasibility, and establishing success criteria that go beyond traditional measures of scope, cost, and time, extending toward business impact, adoption, satisfaction, and

the incorporation of ethical and security standards. Furthermore, it emphasizes defining the team's skills and capabilities from the start to ensure alignment with both technical and business goals (Petrin et al., 2025).

CRISP-DM has historically positioned business understanding as the cornerstone of analytics projects, recommending that project objectives and requirements be clarified in business terms and then translated into technical data mining formulations. In this phase, the methodology suggests assessing the situation in terms of resources, assumptions, risks, costs, and benefits, followed by defining success criteria that capture both business and analytical perspectives. The outcome is a preliminary project plan that aligns stakeholders' expectations and sets the foundation for subsequent steps (Montoya-Murillo et al., 2025; Plotnikova et al., 2020; Wirth & Hipp, 2000). Complementary approaches emphasize the importance of contextual analysis, in which project motivation and stakeholder needs are clarified and feasibility is evaluated. For example, the use of project charters, detailed timelines, and explicit project plans ensures that assumptions, constraints, and intended steps are visible to all parties involved (Bemthuis et al., 2025). Agile-inspired frameworks introduce practices such as Sprint 0, during which objectives, roles, and backlogs are defined through user stories, creativity workshops, and integration sessions. These practices foster early collaboration and adaptability, helping teams refine priorities as understanding evolves (Hamdani et al., 2024; Hayat Suhendar & Widyani, 2023).

The DataPro methodology formalizes the duality between business and technical perspectives by recommending that stakeholders articulate strategic goals and expected outcomes. In contrast, technical teams translate these into precise data-science formulations of inputs, outputs, constraints, and quantitative evaluation criteria. This dual approach reduces ambiguity, enhances communication, and ensures that both dimensions progress in parallel rather than in isolation (Ma et al., 2025). In institutional or academic contexts, scope definition often means aligning project objectives with key performance indicators relevant to education, such as retention rates, graduation rates, and satisfaction levels. Visualization tools like Bizagi BPM may be used to clarify dependencies and bottlenecks, embedding institutional strategy within the scope phase (Azeroual et al., 2025; Herrera Franco et al., 2025). Scientific methodologies, on the other hand, highlight that the scope must address the gap between large-scale data collection and the limited capacity for human analysis. For this reason, they propose framing KDD as an interface that automates large-scale data-handling tasks, enabling scientists to devote their efforts to hypothesis generation and theoretical interpretation. This implies that the scope must explicitly include provisions for scalable infrastructure to process high-dimensional data environments (Fayyad et al., 2000).

Enhanced versions of CRISP-DM advance this discussion by extending the scope definition beyond purely technical or managerial objectives. They include considerations such as value potential across economic, social, and environmental dimensions; human-machine interaction factors, such as ethics, trust, and engagement; and an explicit evaluation of whether AI is the appropriate solution. In addition, they recommend designing early conceptual life cycles of the AI solution, even if incomplete, to identify hidden maintenance costs and anticipate structural design flaws that could undermine long-term value (Bokrantz et al., 2024). Methodologies with a research orientation argue that scope definition must also ensure that problem identification is justified by business or societal needs, that problems are conceptualized, that constraints are clarified, and that objectives are aligned with both quantitative and qualitative measures of success. This iterative refinement between objectives and data insights avoids ex-post justification and reinforces relevance and feasibility (Zolbanin & Aubert, 2025).

In all methodologies, scope definition is not treated as a static, one-time event but rather as an iterative, evolving process. Objectives may be refined based on new insights or changing stakeholder needs, ensuring that analytical initiatives remain relevant and actionable over time (Fois et al., 2020; Hayat Suhendar & Widyani, 2023). This dynamic view underscores the importance of continuous engagement, transparent communication, and adaptive planning tools that sustain alignment between project direction and organizational strategy. Thus, across DMME, CRISP-DM, MAISTRO, DataPro, and other frameworks, the unified guideline for this phase requires that analytics projects establish, from the outset, a shared foundation where business priorities, technical feasibility, ethical foresight, and stakeholder alignment converge. Only through this comprehensive scoping process can later stages of data understanding, preparation, modeling, and deployment achieve their intended impact without losing sight of broader organizational objectives. In sum, scope definition emerges as a multidimensional phase that integrates strategic intent, operational reality, scientific rigor, and ethical considerations, creating the structured basis upon which the sustainability and success of business analytics projects depend.

4.2 Data understanding

In this phase, data understanding is positioned as a central activity that bridges project scope and data preparation, ensuring that available data meaningfully support the project's analytical objectives. It integrates business intent,

technical feasibility, and domain knowledge, serving as the foundation for all subsequent stages of an analytics project. It begins with the systematic collection of raw data from diverse sources, whether originating in organizational systems, sensors, logs, social platforms, or academic repositories, and proceeds to their description, exploration, and validation (Azeroual et al., 2025; Bemthuis et al., 2025; Bhardwaj, 2020; Hamdani et al., 2024; Huber et al., 2019; Wirth & Hipp, 2000). The exploration process requires generating descriptive reports that characterize distributions, anomalies, correlations, and structures in the data while documenting initial hypotheses about hidden information or relevant variables that could influence outcomes (Angée et al., 2018; Azeroual et al., 2025; Montoya-Murillo et al., 2025). Structured frameworks such as the 6Vs model are frequently cited for assessing data quality across dimensions of completeness, accuracy, timeliness, consistency, and representativeness, helping anticipate challenges and confirm dataset adequacy (Ma et al., 2025; Montoya-Murillo et al., 2025; Salazar-Salazar et al., 2024). By combining empirical findings with theoretical models or domain knowledge, practitioners ensure that the exploratory insights not only reveal statistical properties but also contextual meaning, thereby strengthening the alignment between available data and business-driven goals (Angée et al., 2018; Bokrantz et al., 2024; Herrera Franco et al., 2025; Huber et al., 2019; Petrin et al., 2025).

A critical recommendation is that data understanding is not a linear task but rather an iterative, cyclical process that requires ongoing refinement in response to project needs and evaluation feedback (Bokrantz et al., 2024; Petrin et al., 2025; Plotnikova et al., 2020; Salazar-Salazar et al., 2024). For instance, detecting missing values, duplicate records, or inconsistencies may prompt additional data acquisition or preprocessing before modeling, ensuring that data adequacy evolves in tandem with analytical objectives (Bemthuis et al., 2025; Bhardwaj, 2020; Fayyad et al., 2000; Ma et al., 2025; Montoya-Murillo et al., 2025). This adaptability underscores that data understanding evolves alongside project scope, with each cycle of exploration offering the opportunity to refine objectives, adjust hypotheses, and improve data adequacy (Hayat Suhendar & Widayani, 2023; Petrin et al., 2025; Salazar-Salazar et al., 2024; Zolbanin & Aubert, 2025). Moreover, methodologies consistently highlight the importance of linking business needs with data exploration by ensuring that identified variables, subsets, and features remain relevant to operational decisions, strategic goals, or scientific inquiry (Hamdani et al., 2024; Herrera Franco et al., 2025; Petrin et al., 2025; Wirth & Hipp, 2000). This interplay not only validates assumptions but also helps anticipate risks, ensuring that the project maintains focus on its intended outcomes even as technical details evolve.

Domain expertise plays a decisive role during this phase, as practitioners must interpret raw measurements, signals, or categorical indicators within their operational or scientific context (Fayyad et al., 2000; Hamdani et al., 2024; Herrera Franco et al., 2025; Huber et al., 2019). Engineering knowledge, for example, enables the transformation of vibration signals into features that reveal component deterioration, while familiarity with academic processes enables the identification of dropout risk factors in educational data (Herrera Franco et al., 2025; Huber et al., 2019). Such expertise is equally essential when handling heterogeneous datasets—ranging from time-series logs and text streams to genetic sequences and images—where the challenge lies in extracting attributes that meaningfully represent underlying phenomena rather than noise (Bemthuis et al., 2025; Bhardwaj, 2020; Bokrantz et al., 2024; Fayyad et al., 2000). Feature extraction and representation, therefore, become core practices in this stage, ensuring that subsequent modeling is built on relevant and interpretable variables (Fayyad et al., 2000; Huber et al., 2019). These activities not only facilitate technical readiness but also foster a shared understanding across interdisciplinary teams, particularly in projects that require bridging perspectives from different fields (Angée et al., 2018; Bokrantz et al., 2024; Hamdani et al., 2024; Herrera Franco et al., 2025).

At the organizational level, data understanding also requires attention to compliance, access, and governance, given that datasets are often generated across distributed systems and may be subject to regulatory constraints or user-specific contexts (Bokrantz et al., 2024; Hamdani et al., 2024; Zolbanin & Aubert, 2025). Addressing these factors ensures that exploration remains not only technically rigorous but also ethically and legally sound, aligned with institutional governance requirements (Hamdani et al., 2024; Zolbanin & Aubert, 2025). Practical outcomes of this phase include formal documentation such as data description reports, exploration reports, and quality assessment reports, which serve as artifacts to support transparency and reproducibility (Fois et al., 2020; Wirth & Hipp, 2000). Additionally, exploratory outputs such as statistical summaries, visualizations, and descriptive dashboards serve as intermediate deliverables that facilitate communication with stakeholders, enabling iterative validation of assumptions and early alignment between technical findings and managerial expectations (Hayat Suhendar & Widayani, 2023; Herrera Franco et al., 2025; Wirth & Hipp, 2000; Zolbanin & Aubert, 2025).

Ultimately, the phase of data understanding serves as both a diagnostic and constructive step: diagnostic, in that it identifies anomalies, limitations, and risks that could undermine modeling, and constructive, in that it generates preliminary hypotheses, validates feasibility, and defines the operational meaning of data (Angée et al., 2018; Huber et al., 2019; Ma et al., 2025). By establishing a continuous cycle of exploration, evaluation, and refinement, this phase

provides the cognitive and empirical backbone of analytics projects, ensuring that subsequent phases—data preparation, modeling, and deployment—are grounded in reliable, interpretable, and contextually meaningful data (Bokrantz et al., 2024; Petrin et al., 2025; Plotnikova et al., 2020; Salazar-Salazar et al., 2024).

4.3 Data preparation

In this phase, the focus is on systematically transforming raw, heterogeneous, and often noisy data into structured, high-quality, and analyzable datasets that can support reliable modeling and robust decision-making. Serving as a bridge between exploratory understanding and analytical modeling, this phase consolidates exploratory insights from data understanding and operationalizes them through systematic acquisition, cleaning, integration, and feature engineering. It ensures that preparatory work remains aligned with business intent, technical feasibility, and domain knowledge. The process begins with rigorous data acquisition, where information is collected from diverse internal and external sources, including sensors, logs, transactional records, or textual content, and structured using standardized designs that ensure representativeness and analytical validity (Huber et al., 2019). From the outset, explicit documentation of procedures, data quality, limitations, and potential errors is emphasized as a mechanism to secure transparency and accountability throughout the preparation cycle (Huber et al., 2019). Once acquired, data undergoes cleaning and depuration processes that include managing missing values, correcting inconsistencies, detecting and addressing outliers, anonymizing sensitive attributes, and scrubbing noisy records, such as unfiltered social media content. These operations are indispensable for preventing distortions that could compromise subsequent analyses (Angée et al., 2018; Bhardwaj, 2020; Herrera Franco et al., 2025). Textual and unstructured data require specific transformations, such as tokenization, stemming, lemmatization, stopword filtering, or balancing textual classes to ensure comparability and avoid bias. In contrast, numerical datasets require normalization, rescaling, or redundancy reduction (Azeroual et al., 2025). Across both structured and unstructured domains, feature engineering plays a central role: practitioners construct derived variables, select indicators relevant to business and domain objectives, and encode meaningful representations that capture latent patterns while discarding noise (Bokrantz et al., 2024; Ma et al., 2025; Wirth & Hipp, 2000; Zolbanin & Aubert, 2025).

The methodology consistently portrays data preparation as iterative, adaptive, and highly path-dependent. Tasks such as cleaning, integration, transformation, and feature construction are rarely linear; instead, they are repeated multiple times as evaluation feedback or sprint-based development identifies inadequacies that must be addressed (Montoya-Murillo et al., 2025; Petrin et al., 2025; Plotnikova et al., 2020; Wirth & Hipp, 2000). Agile and hybrid frameworks reinforce this adaptability by aligning preparation cycles with sprint iterations and parallel modeling activities, preventing bottlenecks and ensuring that data processing evolves alongside project needs (Hamdani et al., 2024; Hayat Suhendar & Widyani, 2023). Iteration is not only technical but also conceptual: initial design decisions regarding feature labeling, sampling strategies, or integration approaches shape the scope of subsequent preparation activities. They may amplify downstream complexity if handled inadequately (Bokrantz et al., 2024). This path dependency makes early documentation, rationalization of choices, and explicit evaluation of performance metrics crucial for preserving analytical rigor and operational feasibility (Bokrantz et al., 2024). The literature further highlights the importance of dimensionality reduction, whether through projection, segmentation, or partitioning, to generate compact and computationally manageable datasets without sacrificing representational power (Angée et al., 2018; Fayyad et al., 2000; Zolbanin & Aubert, 2025). Concrete applications, such as transforming astronomical images into object descriptors, segmenting DNA sequences into functional regions, or reformatting raw simulation logs into process-mining event logs, illustrate how reduction and structuring preserve relevant information while enhancing interpretability (Bemthuis et al., 2025; Fayyad et al., 2000).

Integration emerges as another recurring principle, ensuring that data from multiple and often inconsistent sources is harmonized, reconciled, and consolidated into unified structures. Tasks include merging datasets, aligning formats, standardizing scales, and ensuring comparability across variables and sites (Hamdani et al., 2024; Ma et al., 2025; Wirth & Hipp, 2000; Zolbanin & Aubert, 2025).

This integration not only addresses technical incompatibilities but also mitigates risks of fragmentation and duplication, which would otherwise reduce trust in the results. Secure environments and automated pipelines are recommended enablers for this integration, enabling scalability, efficiency, and consistency in projects that involve continuous or real-time inputs (Fois et al., 2020; Herrera Franco et al., 2025).

The guidelines consistently underline that the preparation phase is resource-intensive and requires robust infrastructure, particularly when dealing with large-scale or high-dimensional datasets (Bemthuis et al., 2025; Herrera Franco et al., 2025; Petrin et al., 2025). Technical environments that ensure security, scalability, and systematic

processing are considered prerequisites for not only high-quality data but also compliance with organizational and regulatory standards (Fois et al., 2020; Herrera Franco et al., 2025).

Another critical dimension of preparation is the balance between automation and human expertise. Automated reduction, cataloging, or imputation procedures accelerate preprocessing and improve scalability, yet domain expertise remains essential for feature selection, labeling, and interpretation of context-dependent signals (Bokrantz et al., 2024; Fayyad et al., 2000; Zolbanin & Aubert, 2025). Domain knowledge ensures that derived variables are not only statistically meaningful but also aligned with operational and business objectives, strengthening the interpretability and applicability of analytical outcomes (Bokrantz et al., 2024). This dual reliance underscores that preparation is both a technical and an organizational activity, requiring collaboration among data scientists, engineers, and domain specialists (Bokrantz et al., 2024; Ma et al., 2025). Recommendations stress the need for continuous feedback loops: preparation decisions must be validated against modeling requirements and business constraints, with failures or inefficiencies in later phases feeding back into refinement cycles (Hayat Suhendar & Widyani, 2023; Petrin et al., 2025; Zolbanin & Aubert, 2025). Such bidirectional flows guarantee that datasets are not only technically sound but also practically relevant, enabling the models trained on them to generate actionable insights (Hayat Suhendar & Widyani, 2023; Ma et al., 2025).

Deliverables of this phase extend beyond the dataset itself. The literature recommends producing detailed preparation reports documenting acquisition, cleaning activities, feature construction, and integration logic. These artifacts reinforce transparency, reproducibility, and organizational learning while providing a foundation for auditing, compliance, and governance processes. Structured pipelines and technical environments further support repeatability, allowing future updates or dataset extensions without re-engineering the entire preparation workflow (Fois et al., 2020; Salazar-Salazar et al., 2024; Wirth & Hipp, 2000). Far from being a static stage, data preparation is conceived as a living, evolving process that spans the entire project lifecycle, tightly linked to both initial scoping and downstream modeling (Hayat Suhendar & Widyani, 2023; Montoya-Murillo et al., 2025; Wirth & Hipp, 2000). Properly executed, it delivers datasets that are representative, consistent, interpretable, and computationally efficient, while safeguarding security, scalability, and compliance requirements (Fois et al., 2020; Herrera Franco et al., 2025).

Overall, this phase emerges as the cornerstone of business analytics projects, characterized by its iterative nature, reliance on domain expertise, and strong emphasis on technical rigor. It transforms raw, fragmented, and heterogeneous inputs into structured, analyzable datasets that support robust modeling and trustworthy decision-making. Minor errors or omissions at this stage are shown to propagate through the lifecycle, amplifying operational complexity and undermining analytical outcomes (Bokrantz et al., 2024). Conversely, well-designed preparation processes create high-quality datasets that serve as key enablers of effective modeling and reliable insights (Wirth & Hipp, 2000; Zolbanin & Aubert, 2025). For this reason, the literature uniformly underscores that preparation must be executed with systematic discipline, iterative refinement, and organizational commitment, ensuring that the datasets generated are not only valid for analytical purposes but also aligned with the strategic objectives and practical constraints of the business context (Fois et al., 2020; Hamdani et al., 2024; Hayat Suhendar & Widyani, 2023; Petrin et al., 2025).

4.4 Project Design

In this phase, the emphasis is on modeling as the central development activity, with conceptual objectives translated into tangible, testable, and progressively refined analytical artifacts. Building on the structured datasets generated in previous phases, modeling is both an analytical and creative process that guides the transformation of theoretical objectives into practical solutions, such as predictive models, classification systems, and analytical frameworks. The process begins with the design of test structures that guide systematic assessment, enabling the generation of multiple candidate models whose assumptions, parameter settings, and required data formats must be critically evaluated (Fois et al., 2020; Salazar-Salazar et al., 2024). A bidirectional interaction with data understanding and preparation is consistently emphasized, as insights derived from modeling often reveal inconsistencies, limitations, or opportunities for data enhancement (Hayat Suhendar & Widyani, 2023; Montoya-Murillo et al., 2025; Wirth & Hipp, 2000). This cyclical design ensures that refinements are not only technical adjustments but also contribute to the coherence and quality of the overall solution. Iteration across sprints and agile cycles plays a crucial role, as the process is inherently adaptive and oriented toward incremental improvement, with prototypes and partial deliverables serving as tangible checkpoints of progress (Hamdani et al., 2024; Hayat Suhendar & Widyani, 2023; Petrin et al., 2025).

The methodologies converge in underscoring the need to apply multiple statistical, computational, and machine learning techniques, along with parameter calibration, dimensionality reduction, and feature engineering, to enhance

accuracy and interpretability (Azeroual et al., 2025; Bhardwaj, 2020; Plotnikova et al., 2020; Salazar-Salazar et al., 2024). Refinement is achieved through hyperparameter optimization, comparative evaluations across techniques, and the use of both structured and unstructured features that expand the models' explanatory and predictive capacity (Azeroual et al., 2025; Zolbanin & Aubert, 2025). Modeling is portrayed not as a linear process but as a recursive and adaptive one, often requiring revisiting previous steps when discrepancies arise between empirical behavior and theoretical expectations. This adaptability extends beyond technical challenges to address organizational and operational constraints (Bokrantz et al., 2024; Huber et al., 2019).

A distinctive guideline across methodologies is the incorporation of multidisciplinary collaboration and domain knowledge into the iterative refinement cycles. Effective modeling is characterized as a co-creative process that integrates domain expertise, stakeholder feedback, and strategic alignment to ensure contextual relevance and business applicability (Bemthuis et al., 2025; Fayyad et al., 2000; Herrera Franco et al., 2025; Petrin et al., 2025). In particular, the use of process mining, expert validation, and domain-driven interpretation emphasizes that modeling must account for complexity, manage outliers, and balance behavioral coverage with performance indicators (Bemthuis et al., 2025; Herrera Franco et al., 2025). Such integration guarantees that the resulting artifacts are not only accurate but also robust, adaptable, and aligned with the realities of their intended environments (Bokrantz et al., 2024; Huber et al., 2019).

Another recurrent recommendation is to include implementation readiness in the modeling phase. Methodologies increasingly link modeling with deployment by incorporating monitoring mechanisms, scalability strategies, and adaptive feedback loops that ensure models remain reliable under changing real-world conditions (Ma et al., 2025). This bridging perspective reinforces the view that modeling must not be regarded as an isolated research exercise but as an engineering discipline capable of delivering solutions that integrate into operational pipelines, remain adaptable to new information vectors, and sustain performance over time (Huber et al., 2019; Ma et al., 2025). The iterative development of minimal viable models and prototypes demonstrates the capacity to respond quickly to misalignments, to refine dashboards or system outputs, and to sustain user-centered design that prioritizes responsiveness and interpretability (Hamdani et al., 2024; Herrera Franco et al., 2025).

The iterative experimentation cycle also emphasizes balancing innovation with feasibility. Creative and theoretical exploration must coexist with practical considerations such as computational efficiency, operational scalability, and acceptable risk thresholds, ensuring that innovation remains feasible within project constraints (Bokrantz et al., 2024; Zolbanin & Aubert, 2025). Iterative development cycles thus become spaces for negotiation among accuracy, scalability, interpretability, and practicality, ensuring that final artifacts are scientifically valid while operationally sustainable (Bokrantz et al., 2024; Huber et al., 2019; Zolbanin & Aubert, 2025). Deliverables from this phase extend beyond technical outputs to include structured documentation, refinement logs, and knowledge-transfer guidelines that promote traceability, transparency, and institutional learning (Fois et al., 2020; Wirth & Hipp, 2000).

Ultimately, the methodological synthesis highlights modeling as a recursive, collaborative, and adaptive process that transforms abstract objectives into validated, deployable artifacts through cycles of iterative experimentation.

By combining multiple algorithms, testing strategies, expert evaluations, and stakeholder feedback, the phase ensures that models are not static products but evolving solutions that address organizational challenges in real time. By integrating algorithmic diversity, expert validation, and stakeholder feedback, this phase ensures that analytical models evolve as dynamic, context-aware systems capable of continuous improvement. This unified guideline underscores that effective modeling in business analytics projects requires constant refinement, integration of domain knowledge, and engineering foresight to guarantee robustness, adaptability, and long-term value (Azeroual et al., 2025; Hamdani et al., 2024; Huber et al., 2019).

4.5 Model Evaluation

In this phase, model evaluation is conceived as a pivotal quality and alignment checkpoint that bridges model development and deployment, ensuring developed models, insights, and artifacts are not only technically sound but also relevant, actionable, and aligned with organizational objectives. Evaluation is understood as a rigorous, multi-layered, and iterative process embedded throughout the project lifecycle rather than limited to its conclusion. It begins by assessing model performance against predefined technical and business success criteria, including accuracy, precision, recall, latency, operational efficiency, adoption rates, and user satisfaction (Bokrantz et al., 2024; Ma et al., 2025; Petrin et al., 2025; Salazar-Salazar et al., 2024). Models are validated using new or unseen data, alternative samples, or resampling techniques to ensure robustness and generalizability, thereby mitigating risks of overfitting or premature deployment (Angée et al., 2018; Azeroual et al., 2025; Huber et al., 2019; Plotnikova et al., 2020). Beyond technical quality, this phase emphasizes ensuring that outcomes remain consistent with business objectives and

operational requirements, meaning that deployment decisions are based on both predictive accuracy and the capacity to generate measurable business value (Bokrantz et al., 2024; Ma et al., 2025; Montoya-Murillo et al., 2025; Salazar-Salazar et al., 2024; Wirth & Hipp, 2000). Iterative feedback loops are integral to the evaluation process: results may necessitate revisiting earlier phases such as data preparation, modeling, or even objective definition, thereby enabling continuous refinement and adaptation (Bemthuis et al., 2025; Hamdani et al., 2024; Hayat Suhendar & Widyani, 2023; Petrin et al., 2025; Zolbanin & Aubert, 2025).

Evaluation further incorporates systematic comparisons against benchmarks, diagnostic measures, and domain-specific ground truths. These include tests across languages in text analytics, validation against scientific observations in seismology or bioinformatics, and comparisons with institutional outcomes in educational analytics, ensuring both methodological rigor and domain relevance (Bhardwaj, 2020; Fayyad et al., 2000; Herrera Franco et al., 2025). Such practices guarantee that evaluation maintains both methodological rigor and domain relevance, especially in contexts where scientific credibility demands confidence levels exceeding 90% (Fayyad et al., 2000). Likewise, the evaluation process integrates expert judgment and stakeholder validation, where results are scrutinized for plausibility, reliability, and practical applicability before broader adoption (Bemthuis et al., 2025; Fois et al., 2020; Hayat Suhendar & Widyani, 2023; Salazar-Salazar et al., 2024). This dual perspective—technical soundness and domain or business relevance—ensures that models progress only if they satisfy both dimensions.

A distinguishing feature of this phase is its iterative and cyclical nature. Evaluation checkpoints are integrated into project governance activities—such as sprint reviews, stakeholder meetings, and prototype testing—to identify misalignments early, sustain engagement, and guide incremental improvement (Hamdani et al., 2024; Hayat Suhendar & Widyani, 2023). These checkpoints allow early identification of misalignments, reinforce stakeholder engagement, and promote the incremental evolution of models and tools toward business impact (Hamdani et al., 2024; Hayat Suhendar & Widyani, 2023). Furthermore, evaluation explicitly includes trade-off analysis, balancing criteria such as accuracy versus interpretability, computational cost versus scalability, or predictive performance versus usability, ensuring that chosen solutions remain both robust and feasible in production contexts (Bokrantz et al., 2024; Zolbanin & Aubert, 2025). Complementary validation processes—including user acceptance testing and system verification—provide safeguards that ensure operational readiness and compliance with institutional or regulatory standards before deployment (Fois et al., 2020).

The methodological novelty of this phase lies in combining quantitative validation with qualitative and organizational assessments. Metrics such as AUC, F1-score, Precision-Recall, and Matthews Correlation Coefficient provide statistical rigor (Azeroual et al., 2025; Zolbanin & Aubert, 2025), while structured reviews, expert assessments, and stakeholder acceptance confirm interpretability, trust, and alignment with organizational goals (Bemthuis et al., 2025; Hayat Suhendar & Widyani, 2023; Salazar-Salazar et al., 2024). Deliverables emerging from this phase include evaluation reports, documentation of process reviews, and lists of possible subsequent actions, which function as decision points for advancing or refining models (Fois et al., 2020; Wirth & Hipp, 2000). Overall, evaluation is framed not as a terminal checkpoint but as a continuous mechanism of quality assurance, business alignment, and organizational learning that bridges analytical development with deployment readiness. By systematically integrating technical metrics, business indicators, domain-specific benchmarks, and iterative stakeholder feedback, this phase ensures that business analytics projects deliver statistically valid outcomes, operationally robust, and strategically valuable (Montoya-Murillo et al., 2025; Salazar-Salazar et al., 2024; Wirth & Hipp, 2000; Zolbanin & Aubert, 2025).

4.6 Governance and Ethics

In this phase, governance and ethics operate as transversal and foundational dimensions that permeate the entire analytics project lifecycle, operating in parallel with the technical evaluation of models to ensure that results are not only accurate but also aligned with organizational, legal, and cultural values. Rather than a discrete or final step, governance functions as a continuous safeguard that ensures analytical outputs remain technically sound while consistent with legal, organizational, and societal standards. This cross-cutting integration reinforces that governance is inseparable from analytics execution, guiding decisions from project initiation to long-term maintenance. This dedicated phase ensures that analytics projects adhere to evolving AI governance standards by institutionalizing traceability, transparency, and accountability throughout the lifecycle—principles emphasized in the Ethics Guidelines for Trustworthy AI issued by the European Commission’s High-Level Expert Group on AI (2019). Embedding these principles into every phase of analytics practice transforms governance and ethics into proactive design components that prevent bias, opacity, and noncompliance from arising as afterthoughts, fostering responsible innovation and public trust. This inclusion aligns with international AI governance standards emphasizing traceability

and accountability, such as the European Commission's Trustworthy AI Guidelines (High-Level Expert Group on Artificial Intelligence, 2019). Evaluation requires that models are validated against both business and technical objectives, using robust procedures to guarantee predictive reliability and actionable insights under real operational conditions (Huber et al., 2019). This validation is complemented by governance structures that provide role clarity, stakeholder accountability, and transparent documentation of assumptions and processes, thereby reinforcing trust in the outputs (Hamdani et al., 2024; Herrera Franco et al., 2025). Governance is not conceived as a separate or late-stage activity but rather as a cross-functional integration of experts, continuous feedback loops, and transparent communication mechanisms that reduce ambiguity and prevent competence bottlenecks (Bokrantz et al., 2024). Constant communication among experts, stakeholders, and decision-makers ensures alignment, reduces ambiguity, and reinforces accountability across the analytics lifecycle. Ethical principles are embedded from the very beginning of projects, with emphasis on compliance with national and international regulations such as LGPD, adherence to confidentiality rules, and the anonymization of sensitive data (Azeroual et al., 2025; Petrin et al., 2025). Transparency is further strengthened through dashboards, methodological documentation, reproducibility practices, and open access where possible, ensuring that stakeholders across departments can validate outcomes and build confidence in the integrity of results (Hamdani et al., 2024; Herrera Franco et al., 2025). Fairness and accountability emerge as recurring checkpoints, requiring explicit mechanisms for bias mitigation, the inclusion of diverse datasets, and the establishment of criteria that avoid arbitrary thresholds in variable selection (Azeroual et al., 2025; Herrera Franco et al., 2025). These safeguards ensure that models remain valid, replicable, and culturally appropriate across heterogeneous contexts, respecting local practices while maintaining global standards of responsible data use (Azeroual et al., 2025; Petrin et al., 2025). The methodology also stresses the importance of iterative communication with stakeholders to foster a culture of organizational learning and to integrate ethical considerations into daily decision-making, rather than treating them as isolated verifications (Bokrantz et al., 2024; Hamdani et al., 2024). This orientation reframes governance and ethics from compliance requirements into strategic enablers of sustainable innovation, directly influencing the usability, scalability, and credibility of analytic solutions. The integration of monitoring mechanisms, transparency checks, and user feedback into the evaluation process ensures that models remain resilient to changes in data quality dimensions, such as completeness, plausibility, timeliness, and consistency (Azeroual et al., 2025). At the same time, ethical vigilance mitigates reputational risks arising from neglecting fairness or privacy safeguards (Azeroual et al., 2025). While some methodologies mention ethics only briefly in the data understanding phase, without structured recommendations (Zolbanin & Aubert, 2025), the convergence of CRISP-DM extensions, MAISTRO, and other frameworks demonstrates a clear shift toward embedding governance and ethics as ongoing, systemic practices, inseparable from technical validation (Bokrantz et al., 2024; Petrin et al., 2025). Ultimately, this phase reframes evaluation as a holistic assurance process—where transparency, compliance, accountability, and fairness are continuously monitored and operationalized across all stages of analytics execution—thereby aligning analytical systems with both strategic business objectives and the ethical responsibilities of the organization (Azeroual et al., 2025; Bokrantz et al., 2024; Hamdani et al., 2024; Zolbanin & Aubert, 2025).

4.7 Operation and Maintenance

In this phase, operation and maintenance are reconceptualized so that deployment is understood not as a final step but as the beginning of an ongoing operational lifecycle in which sustainability, adaptability, and monitoring are central to ensuring that analytical solutions continue to deliver value. Frameworks such as DMME emphasize the importance of connecting models to runtime-capable infrastructures and providing long-term resilience in environments with continuous data flows (Huber et al., 2019). Similarly, MAISTRO advances a structured perspective on operationalization, emphasizing version management, retraining cycles, and constant monitoring as mechanisms of adaptability and sustained performance (Petrin et al., 2025). Extensions of CRISP-DM formalize these principles by including explicit tracking, updating, and sustaining tasks, and by recommending automation, retraining policies, and human-in-the-loop mechanisms to maintain trust and relevance over time (Bokrantz et al., 2024). The convergence of these recommendations indicates that successful deployment requires integrating predictive and prescriptive outputs directly into organizational decision workflows, thereby embedding analytical tools into daily operations rather than treating them as stand-alone artifacts (Azeroual et al., 2025; Montoya-Murillo et al., 2025).

Monitoring emerges as the guiding principle of the phase, with methodologies such as TDSP and DDSL underscoring dashboards, system health indicators, and periodic supervision as core deliverables (Salazar-Salazar et al., 2024). These tools ensure that deployed solutions remain aligned with strategic objectives, provide real-time visibility, and reduce reliance on static reports (Hamdani et al., 2024). In both industrial and institutional contexts, dashboards and predictive alerts are used to continuously track indicators—ranging from production alignment to

academic performance—facilitating proactive interventions and reducing delays in corrective action (Herrera Franco et al., 2025). The operational challenge is not only technological but organizational, as methodologies acknowledge the presence of political, cultural, and economic complexities that condition the adoption and sustainability of deployed solutions (Bemthuis et al., 2025). This is particularly relevant in data-intensive domains that require scalability and computational efficiency, where adaptive infrastructures and high-performance environments ensure analytical continuity amid growing data volumes (Fayyad et al., 2000).

Sustainability in deployment thus implies a dual responsibility: ensuring the robustness of technical infrastructures and aligning ongoing operation with institutional governance. Recommendations call for sustained retraining, drift detection, and incremental model refinement to ensure predictive tools remain accurate as environments shift (Bokrantz et al., 2024). These processes extend to managing production systems, overseeing infrastructure, documenting procedures, and optimizing resources, while also ensuring security and defect management (Fois et al., 2020). Governance emerges as a transversal layer, requiring policies for time-based or condition-based maintenance of data pipelines and deployed models (Bokrantz et al., 2024), complemented by help-desk administration and user support communities that reinforce adoption and trust (Fois et al., 2020). Through this dual perspective—technical and institutional—operation becomes a living, governance-driven process of sustainability that embeds analytics into the organization’s daily operational culture. Far from a purely technical exercise, deployment involves integrating analytical insights into existing institutional routines, whether through supply-chain coordination (Hamdani et al., 2024), academic management processes (Herrera Franco et al., 2025), or research workflows connected to evolving data sources (Azeroual et al., 2025).

At the same time, deployment cannot be detached from the cycle of evaluation and feedback. Case experiences underscore the importance of iterative validation, in which monitoring not only tracks technical performance but also verifies the benefits delivered, thereby informing adjustments in subsequent projects (Wirth & Hipp, 2000; Zolbanin & Aubert, 2025). This feedback-driven continuity ensures that operation and evaluation remain tightly connected, promoting refinement whenever misalignments between model outputs and organizational needs arise. This iterative dimension emphasizes that operationalization must remain open to revisiting earlier project phases when models fail to meet business or domain criteria (Zolbanin & Aubert, 2025). Maintenance is therefore framed as both a corrective and generative activity, enabling continuous learning and the evolution of practices across projects. Where methodologies fail to provide explicit post-deployment recommendations—as seen in some research-oriented approaches (Hayat Suhendar & Widyani, 2023)—this absence underscores the growing need for systematic operational guidance that extends beyond initial implementation into the long-term management of AI systems.

Ultimately, this phase represents a methodological consolidation of deployment and operation as inseparable components of business analytics projects. It integrates infrastructure design (Huber et al., 2019), structured operational practices (Petrin et al., 2025), monitoring plans (Wirth & Hipp, 2000), governance and complexity awareness (Bemthuis et al., 2025), and dashboards for real-time supervision (Hamdani et al., 2024; Herrera Franco et al., 2025; Salazar-Salazar et al., 2024). It also incorporates domain-specific sustainability concerns such as scalability in data-intensive contexts (Fayyad et al., 2000), institutional embedding of predictive outputs (Azeroual et al., 2025; Montoya-Murillo et al., 2025), and explicit maintenance policies for drift detection and retraining (Bokrantz et al., 2024; Fois et al., 2020). Taken together, these practices define operation and maintenance as dynamic, adaptive, and governance-oriented processes that sustain analytical reliability, organizational alignment, and long-term business value.

4.8 Continuous Improvement

In this phase, continuous improvement serves as the sustaining engine of the analytics lifecycle, conceived as a cyclical, adaptive mechanism that extends across the entire lifecycle of business analytics projects, ensuring that both methodological processes and analytical solutions remain relevant and effective in dynamic organizational environments. Far from representing closure, this phase embodies the iterative and renewal-oriented spirit of modern analytics practice, where learning and feedback drive the long-term evolution of analytical systems. This phase aligns with emerging MLOps (Machine Learning Operations) practices that emphasize continuous model monitoring, automated retraining, and pipeline updates to sustain solution performance and reliability in production environments. Integrating these practices reinforces the governance-aware perspective of the unified framework, connecting methodological improvement with the operational discipline required for scalable and trustworthy analytics systems. This alignment mirrors the iterative principles of MLOps, which promote retraining, deployment supervision, and adaptive scalability to sustain analytical value. While some methodologies, such as DMME, limit their perspective to iterative refinement during design and evaluation without addressing long-term feedback loops (Huber et al., 2019),

others explicitly embed improvement as a structural principle. CRISP-DM, for instance, positions learning as cyclical, where lessons learned trigger new, more focused business questions, documentation is treated as a “living” asset, and iterations are explicitly planned to incorporate accumulated experience (Bokrantz et al., 2024; Plotnikova et al., 2020; Salazar-Salazar et al., 2024; Wirth & Hipp, 2000). Similarly, frameworks inspired by agile approaches highlight the roles of sprint-based refinement, retrospectives, and feedback loops in reinforcing adaptation, enhancing team skills, and reducing rework (Hamdani et al., 2024; Hayat Suhendar & Widayani, 2023). In the unified framework, this iterative culture extends beyond model optimization to encompass organizational learning, process enhancement, and stakeholder engagement, recognizing that improvement emerges from sustained monitoring, contextual evolution, and feedback integration (Bokrantz et al., 2024; Fois et al., 2020; Salazar-Salazar et al., 2024).

The methodological emphasis is on embedding feedback loops between phases to continuously refine data, models, and evaluations. Some approaches recommend revisiting earlier phases whenever limitations are detected, iteratively refining analytical strategies, or integrating new data sources to improve validity and generalizability (Angée et al., 2018; Azeroual et al., 2025; Bemthuis et al., 2025). Others point to practical examples, such as refining sentiment analysis by repeatedly collecting and cleaning real-time data, or retraining models with larger, more diverse samples to improve robustness (Bhardwaj, 2020). Process mining examples highlight the role of repeated checks for validity, expansion to additional indicators, and involvement of broader expert panels, illustrating that iterative face-validity exercises are essential to sustain rigor (Bemthuis et al., 2025). Across domains, methodologies stress that improvement is not restricted to single problem instances but requires generalization, guided emergence, and adaptation to evolving datasets, technologies, and organizational practices (Azeroual et al., 2025; Fayyad et al., 2000; Zolbanin & Aubert, 2025). Through these mechanisms, continuous improvement operates as both a corrective and generative process, enhancing robustness while fostering institutional learning.

Continuous improvement is also tied to the adaptability and extensibility of systems. Methodologies highlight the importance of dynamically adjusting predictive models and strategies as conditions evolve, recommending replication across contexts, incorporating demographic or socioeconomic variables, and exploring new modalities to sustain relevance (Herrera Franco et al., 2025). Frameworks like MAISTRO emphasize recurring loops across phases and underscore continuous learning not only for models but also for teams, promoting adaptation to new technologies and practices (Petrin et al., 2025). In more technical domains, extensible environments such as SKICAT or JARtool illustrate how methodologies evolve by expanding analytical capabilities, whether through supervised-to-unsupervised extensions or through adaptation to novel image or sequence data (Fayyad et al., 2000). This extensibility orientation ensures long-term sustainability, enabling methodologies to evolve alongside advances in machine learning, data availability, and institutional needs (Azeroual et al., 2025).

Another recurring recommendation is to formalize reflection and learning as an explicit mechanism for improvement. Some methodologies encourage systematic reviews of results, given their capacity to generate new research questions or business insights (Angée et al., 2018). Others incorporate structured reflections through retrospectives or dedicated phases to document lessons learned, conduct internal reviews, and institutionalize design principles that transcend individual projects (Fois et al., 2020; Hayat Suhendar & Widayani, 2023; Zolbanin & Aubert, 2025). In this way, reflection serves both operational and epistemic functions—advancing methodological rigor while enabling theoretical and organizational learning (Zolbanin & Aubert, 2025).

The overall synthesis highlights that continuous improvement requires a dual orientation: on one hand, operational refinement through iterative optimization of data, models, and processes; and on the other, organizational and methodological evolution through documentation, reflection, and adaptation. Thus, improvement is achieved by revisiting prior results, adapting methods to new data and risks, and integrating user feedback and contextual insights into subsequent iterations (Azeroual et al., 2025; Bokrantz et al., 2024; Salazar-Salazar et al., 2024). By embedding feedback as a permanent structural feature, continuous improvement ensures that analytics projects accumulate resilience, accuracy, and institutional knowledge—fostering sustained innovation and adaptive capacity in volatile, data-intensive environments.

4.9 Summary and Synthesis

Collectively, the eight PRODIG8 dimensions capture the methodological evolution of business analytics project execution and reveal three functionally distinct components of its architecture: a core execution lifecycle, transversal control, and adaptive feedback. The six core execution dimensions organize the progression from project definition and data work to model evaluation and operational maintenance; Governance and Ethics provides transversal oversight across this lifecycle; and Continuous Improvement connects project results, operational experience, and environmental change back to the execution dimensions through adaptive feedback. This structure reflects the

evolution of analytics methodologies from predominantly technical, sequential workflows to integrated systems that combine analytical execution with organizational accountability, governance, and continuous learning. PRODIG8 therefore consolidates methodological diversity not as eight equivalent sequential phases, but as an integrated architecture in which execution, control, and adaptation perform distinct but interdependent functions.

5 Discussion

Building on the comparative synthesis presented in Section 4, this discussion interprets the broader implications of the PRODIG8 framework for advancing theory, practice, and governance in analytics project management. The model's contribution lies not only in consolidating methodological content across existing frameworks, but also in differentiating three complementary functions required for analytics-project execution: a core execution lifecycle, transversal governance and ethical control, and adaptive feedback through continuous improvement. This distinction more explicitly represents how technical execution, organizational accountability, and learning interact within the same methodological architecture. Accordingly, PRODIG8 extends conventional lifecycle models such as CRISP-DM, ASUM-DM, and TDSP by integrating execution, control, and adaptation as distinct but interdependent components of analytics-project management.

Unlike prior frameworks that focus on specific organizational or disciplinary settings—such as the conceptual HR analytics model proposed by Gurusinghe et al. (2021)—the unified framework developed here bridges multiple domains and offers methodological guidance applicable across sectors. This broader orientation positions the study as a higher-level generalization of domain-specific approaches, consistent with the cross-disciplinary expectations of contemporary analytics research.

The structural differentiation of PRODIG8 into execution, control, and adaptation extends the theoretical contribution beyond methodological consolidation. From a standardization perspective, the six core execution dimensions establish a common process architecture that reduces procedural variability while preserving methodological interoperability across organizational contexts. From a governance perspective, treating Governance and Ethics as a transversal control dimension recognizes accountability, transparency, risk management, and ethical oversight as continuous requirements rather than activities confined to a particular stage. From a systems-integration perspective, Continuous Improvement provides the feedback mechanism through which execution outcomes, operational experience, and changing conditions can modify activities throughout the lifecycle. PRODIG8 therefore combines process standardization with transversal control and adaptive feedback within a single architecture, providing a theoretical explanation for how analytics methodologies can remain structured without becoming rigid.

The findings reveal a progressive methodological evolution from analytics methodologies centered primarily on the technical execution workflow toward broader socio-technical architectures incorporating governance, operationalization, and continuous learning. Early approaches such as KDD, SEMMA, and CRISP-DM established the core logic for transforming business or analytical objectives and data into evaluated analytical outputs. Subsequent methodologies progressively extended this logic by introducing stronger project-management practices, operational and maintenance requirements, governance and ethical mechanisms, and explicit feedback and learning processes. PRODIG8 captures this cumulative evolution by separating these methodological contributions according to their functional role: execution dimensions structure the development and operationalization of the analytical solution, Governance and Ethics continuously control how these activities are performed, and Continuous Improvement enables the architecture to respond to evaluation results, operational experience, and changing conditions. PRODIG8 therefore contributes not simply by adding activities absent from earlier methodologies, but by integrating methodological advances accumulated across successive generations into a differentiated execution–control–adaptation architecture. In contrast, more recent frameworks—such as DMME, ASUM-DM, TDSP, MAISTRO, and DataPro—incorporate agile principles, feedback loops, and governance mechanisms, marking a clear shift from static, tool-based procedures toward dynamic, learning-oriented systems aligned with strategic and human-centered objectives.

This transition is equally evident in academic governance contexts, where process mining and CRISP-DM-based curriculum analysis frameworks illustrate the integration of data analytics and institutional management under shared governance principles (Herrera Franco et al., 2025). Empirical implementations, such as the agile–CRISP-DM hybrid applied in supply chain analytics (Hamdani et al., 2024), exemplify how iterative cycles foster organizational learning, stakeholder engagement, and adaptive project management within real operational contexts. Within this shift, the unified framework also provides scalable guidance for small and medium-sized enterprises, enabling them to adopt agile analytics lifecycles that preserve methodological rigor while accommodating limited resources and rapidly

changing operational needs. Together, these developments represent the foundation on which the unified framework was constructed.

The PRODIG8 architecture integrates methodological advances that were introduced incrementally across successive generations of analytics-project methodologies. Within the core execution lifecycle, Project Scope Definition extends conventional business-understanding activities by incorporating feasibility, stakeholder alignment, and broader project constraints; Data Understanding and Data Preparation consolidate iterative practices related to data quality, heterogeneous sources, and analytical readiness; and Project Design and Model Evaluation integrate structured development with repeated validation and feedback. Operation and Maintenance extends the execution lifecycle beyond initial deployment by incorporating monitoring, retraining, and long-term operational requirements. Beyond these execution dimensions, Governance and Ethics is represented as a transversal control mechanism that embeds transparency, accountability, and ethical oversight throughout project execution, while Continuous Improvement provides the adaptive mechanism through which evaluation results, operational experience, and organizational learning can trigger refinement across the lifecycle. PRODIG8 therefore integrates methodological advances that are dispersed across earlier frameworks into a differentiated execution–control–adaptation architecture.

Collectively, these refinements — embodied in the PRODIG8 framework — mark a transition from tool-centered, static process models to dynamic, governance-aware systems that align technical execution with ethical and managerial accountability, thereby advancing the methodological maturity of business analytics. The following subsections elaborate on these insights through theoretical, practical, and bridging perspectives. By clarifying roles, incorporating continuous learning, and embedding governance protocols, the framework directly mitigates risk factors like miscommunication and misaligned objectives—two of the most common causes of project failure (Saltz & Krasteva, 2022). To organize the interpretation, Section 5 is divided into three perspectives: theoretical, bridging, and practical.

5.1 Theoretical Implications

From a theoretical standpoint, the proposed framework advances the conceptual architecture of analytics methodologies by merging process standardization with adaptive design principles. It extends classical models such as CRISP-DM and ASUM-DM by embedding governance and continuous improvement as structural elements rather than as auxiliary components. This synthesis aligns with international standards such as ISO/IEC 29110 and IEEE 7000, emphasizing iterative feedback, traceability, and ethical oversight. By linking technical processes with managerial and governance layers, the framework strengthens the theoretical convergence of data science, software engineering, and organizational management—establishing analytics as a mature interdisciplinary domain rather than a collection of isolated practices.

5.2 Bridging Theory to Practice

The theoretical contributions outlined above translate directly into practical value through the framework’s modular and iterative structure. It serves as a bridge between academic rigor and organizational application, demonstrating how governance, adaptability, and transparency can be operationalized across diverse project contexts. Each phase is designed to scale with organizational maturity, regulatory requirements, and technological complexity. By situating governance and feedback mechanisms within every stage of the analytics lifecycle, the framework transforms abstract methodological principles into actionable guidance. This bridging perspective reinforces the continuity between conceptual development and managerial execution, eliminating the traditional divide between research-based methodologies and real-world implementation. The unified framework operationalizes the agile and traceability principles central to MAISTRO by embedding iterative feedback, documentation control, and continuous validation as governance mechanisms applicable beyond AI contexts.

5.3 Practical Implications

Although this study is conceptual in scope, the unified framework could be illustrated through an empirical adaptation similar to Bemthuis et al. (2025), who embedded CRISP-DM phases into the assessment of agent-based simulation models using process mining. Compared to prior domain-specific frameworks—such as the HR analytics model by Gurusinge et al. (2021)—our framework is designed for broader cross-sector applicability. Following this precedent, the proposed governance-aware structure could be applied to validate analytics projects by aligning iterative feedback loops, documentation control, and evaluation mechanisms with real-world data workflows. Such

an example would demonstrate how the framework translates methodological integration into operational practice, reinforcing its empirical relevance and usability.

For practitioners and project sponsors, adopting this unified framework offers tangible managerial benefits. It helps mitigate common failure factors in analytics initiatives by clarifying team roles, improving cross-functional communication, and embedding continuous learning cycles within project governance. In practice, the framework provides a roadmap to avoid trial-and-error and siloed execution, promoting consistency, accountability, and knowledge transfer across teams. By translating methodological rigor into actionable coordination mechanisms, it enables organizations to enhance success rates and reduce the inefficiencies that have historically characterized data science projects (Saltz & Krasteva, 2022).

A comparable opportunity exists by extending the unified framework toward applications consistent with the DataPro model (Ma et al., 2025), which emphasizes the dual alignment between business and technical understanding. For example, the governance-aware structure proposed here could be adapted to operational contexts where roles, responsibilities, and communication flows are explicitly defined between data scientists and stakeholders, as demonstrated in DataPro’s eco-driving case. Such an adaptation would empirically validate the framework’s ability to translate governance and ethical design principles into cross-functional project coordination.

In practice, the framework provides organizations with a structured framework for designing, executing, and sustaining analytics initiatives. Its eight-phase configuration provides precise alignment between analytical tasks, strategic goals, and governance requirements, allowing decision-makers to monitor performance and ensure accountability throughout the project lifecycle. The model also facilitates the creation of maturity assessments, training programs, and audit tools that evaluate transparency, risk management, and compliance. By embedding iterative feedback and ethical oversight into daily workflows, the framework promotes responsiveness to technological change and fosters a culture of continuous improvement. In this way, it operationalizes the conceptual advances discussed earlier, demonstrating how methodological integration enhances both project efficiency and institutional trust. Notably, the Continuous Improvement phase aligns with modern MLOps practices, emphasizing monitoring, retraining, and explainability. Similarly, the framework’s iterative loops reflect principles from agile project management, as practiced in many modern data teams.

5.4 Summary of Implications

In summary, the unified framework consolidates decades of methodological evolution into a single, adaptable structure that strengthens the scientific, managerial, and ethical foundations of analytics project management. By combining theoretical rigor with practical guidance, it enables consistent governance, scalability, and transparency across sectors. The synthesis reinforces the professionalization of analytics practice and provides a foundation for future empirical validation and refinement. These insights also establish a natural transition to the following section, which outlines the study’s limitations and identifies future pathways for validation and methodological advancement. These results provide the foundation for evaluating the study’s scope and constraints, discussed next in Section 6 (Limitations). Having articulated the theoretical, bridging, and practical implications of PRODIG8, the study now turns to examine its limitations and delineate areas requiring empirical validation and refinement.

6 Limitations

The proposed PRODIG8 framework synthesizes decades of methodological development, yet several limitations must be acknowledged. These constraints define priorities for future empirical validation and refinement.

- **Scope limitations:** The analysis relies on a corpus of 18 documents retrieved exclusively from the Scopus database. While Scopus provides extensive coverage, it may exclude relevant methodologies published in non-indexed outlets, industry reports, or proprietary materials. Consequently, the study may not capture all practitioner-oriented or emergent frameworks.
- **Methodological limitations:** PRODIG8 is analytically derived from published methodological frameworks rather than from observations of completed analytics projects. Although the sequential structural-reconciliation procedure and subsequent mapping of methodological recommendations provide a systematic basis for model construction, the resulting architecture has not yet been empirically evaluated in real-world project environments. Moreover, differences in terminology, granularity, and scope across the reviewed methodologies required interpretive judgments to determine conceptual equivalence, split or merge functions, and map recommendations to the final dimensions.

- Temporal limitations: The synthesis reflects the state of the analytics landscape at the time of data collection. It may require revision as the field evolves—particularly with advances in automation, explainable AI, and data-governance mechanisms. For example, the rapid adoption of generative AI and autonomous decision systems may redefine governance expectations.
- Language limitations: The restriction to English-language sources may exclude non-English methodological contributions.
- Analytical subjectivity: Constructing PRODIG8 necessarily involved interpretive decisions during both structural reconciliation and recommendation mapping. Two reviewers made these decisions systematically and independently and then reconciled them through consensus; nevertheless, alternative interpretations of methodological components remain possible. Future research can therefore complement the present analytical derivation through external expert assessment, structured applications to completed analytics projects, and empirical examination of the relationships represented in the model.
- Lack of validation: Although conceptual, future validation efforts could include expert panels, longitudinal case studies, or structured mappings of published projects to the framework’s phases. These steps will ensure practical relevance and improve phase granularity.

Collectively, these limitations highlight the need for empirical validation and longitudinal studies to evaluate the framework’s applicability, scalability, and practical relevance across diverse organizational contexts and industries. They also define priorities for methodological refinement and provide a foundation for the research directions discussed in the following section.

7 Conclusions

Building on the sequential reconciliation and consolidation of 18 analytics-project methodologies, this study develops PRODIG8 as an integrated architecture for the systematic execution, governance, and adaptation of business analytics projects. The model comprises eight interdependent methodological dimensions with three distinct functional roles. Project Scope Definition, Data Understanding, Data Preparation, Project Design, Model Evaluation, and Operation and Maintenance constitute the core execution lifecycle; Governance and Ethics provides transversal control across that lifecycle; and Continuous Improvement provides the adaptive feedback mechanism through which project results, operational experience, stakeholder feedback, and changing conditions can inform preceding activities. This execution–control–adaptation architecture extends methodological integration beyond the consolidation of lifecycle stages by explicitly differentiating the functions through which analytics projects are executed, governed, and progressively refined. PRODIG8 thereby provides a common analytical structure for organizing and examining analytics-project practices across organizational and application contexts.

While conceptually robust, the framework remains theoretical and requires empirical validation. Future research should therefore test and operationalize it in real-world contexts through longitudinal case studies, cross-sector applications, or controlled experiments to assess its impact on project outcomes, scalability, and stakeholder engagement. An additional validation pathway involves educational analytics and curriculum management, where CRISP-DM-based process mining models provide an ideal context to test the unified framework’s adaptability and its governance-oriented structure (Herrera Franco et al., 2025). Additional efforts could develop maturity assessment tools and performance indicators that evaluate organizational capabilities for implementing analytics initiatives in accordance with the proposed phases. Future empirical validation should focus on engineering and sensor-intensive contexts to test how the unified framework operationalizes DMME’s technical phases within real production environments and cyber-physical systems. A promising validation pathway involves hybrid applications that combine CRISP-DM with agile project management approaches, as illustrated by Hamdani et al. (2024), which demonstrate how iterative sprints and stakeholder-driven learning cycles operationalize methodological integration in complex supply chain environments. In particular, further studies should explore how the unified governance-aware framework can function as an agile SDLC model for Big Data Analytics Systems in small and medium-sized organizations, extending the comparative foundations established by Salazar-Salazar et al. (2024) and validating its adaptability under resource constraints.

As analytics ecosystems evolve toward AI-augmented, automated, and explainable systems, continued inquiry should embed continuous improvement, interpretability, and governance mechanisms into adaptive, data-driven environments. Such research will not only strengthen the empirical foundation of the framework but also guide the emergence of future governance-oriented analytics systems that integrate accountability, transparency, and ethical oversight as core design principles.

From a practical standpoint, PRODIG8 provides a structured analytical reference for examining the completeness and coherence of an organization's approach to analytics-project execution. The six core execution dimensions enable practitioners to examine whether activities spanning project definition, data understanding and preparation, analytical design, evaluation, and operational maintenance are adequately represented, while the transversal Governance and Ethics dimension directs attention to accountability, transparency, risk, and ethical oversight throughout those activities. Continuous Improvement complements these functions by requiring project results, operational experience, stakeholder feedback, and changing conditions to be systematically reconsidered across the execution lifecycle. PRODIG8 can therefore be used not only to organize project activities, roles, and deliverables, but also to identify methodological areas that are absent, weakly represented, or insufficiently connected within an existing project approach. In this way, the model provides a common architecture for comparing, reviewing, and refining analytics-project practices while preserving the adaptability required across organizational contexts.

Ultimately, this study lays the groundwork for bridging analytics theory and practice, fostering a mature, standardized, and ethically grounded discipline capable of evolving into the next generation of governance-aware, adaptive analytics systems. In the long term, the proposed framework can inform the development of institutional standards, accreditation guidelines, maturity models, and methodological curricula in analytics education and professional certification, fostering convergence between academic research, industry practice, and governance systems. In this sense, the unified framework offers a solid foundation for developing governance-aware standards that can inform and harmonize the next generation of analytics methodologies. In methodological terms, this progression extends the lineage initiated by DMME, transferring its engineering-focused rigor to a broader governance-aware framework that now encompasses business, institutional, and cross-sector analytics domains. To enhance its empirical robustness, future research should validate the framework within real-world organizational contexts, assessing both implementation outcomes and stakeholder perceptions. By unifying methodological, managerial, and ethical dimensions, this study not only advances analytics methodology but also provides an actionable foundation for evidence-based governance, making it a relevant contribution to both scholarly research and applied data science practice.

In summary, PRODIG8 provides a unifying foundation that connects decades of methodological evolution with contemporary governance and ethical standards. Its adoption can catalyze the transition of analytics from a fragmented set of practices into a mature, standardized discipline—offering both scholars and practitioners a common methodological language for designing, governing, and improving analytics initiatives.

8 References

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